

Time Series Overview



You said:

time series

ChatGPT said:

Time Series (Easy → Advanced Explanation)

◆ 1. What is Time Series?

Time Series is data that is collected **over time at regular intervals**.

👉 Example:

- Daily temperature
- Monthly sales
- Yearly population
- Stock prices every minute

Example:

Code	
Day	Sales
1	100
2	120
3	150
4	130

Here, **time order matters**.

◆ 2. Types of Time Series Components

A time series usually has 4 main components:

1 Trend (T)

Long-term upward or downward movement.

Example:

- Increasing population
- Growing company revenue

 Upward Trend

 Downward Trend

2 Seasonality (S)

Repeating pattern in fixed period (daily, monthly, yearly).

Example:

- Ice cream sales increase every summer
- Festival season shopping increase

3 Cyclical (C)

Long-term fluctuation but not fixed time period.

Example:

- Economic boom and recession cycles

4 Irregular / Noise (R)

Random variation (unexpected events).

Example:

- COVID impact
- Natural disasters

◆ 3. Types of Time Series Models

1 Additive Model

Used when variations are constant.

Formula:

```
Code  
Y = T + S + C + R
```

2 Multiplicative Model

Used when variations increase with trend.

Formula:

```
Code  
Y = T × S × C × R
```

Important Time Series Models

1 AR Model (AutoRegressive)

Current value depends on past values.

```
Code  
Yt = c + φ1Yt-1 + φ2Yt-2 + ...
```

2 MA Model (Moving Average)

Current value depends on past errors.

3 ARMA Model

Combination of AR + MA.

4 ARIMA Model

Full form: **AutoRegressive Integrated Moving Average**

Used when data is **non-stationary**.

5 SARIMA Model

Seasonal ARIMA model.

6 Prophet Model

Developed by **Facebook** (now **Meta Platforms**)

Good for business forecasting.

◆ 4. What is Stationarity?

A time series is stationary when:

- Mean is constant
- Variance is constant
- No trend or seasonality

How to check?

- ADF Test (Augmented Dickey Fuller Test)

If p-value < 0.05 → Stationary

◆ 5. Differencing (To Make Stationary)

```
Code  
Yt' = Yt - Yt-1
```

Removes trend.

◆ 6. ACF and PACF

ACF (AutoCorrelation Function)

Shows correlation with past values.

PACF (Partial AutoCorrelation Function)

Shows direct relationship with lag.

Used to determine:

- AR order (p)

- MA order (q)

7. Time Series in Python (Basic Example)

```
Python
import pandas as pd
import matplotlib.pyplot as plt

data = [100,120,130,150,170,160]
plt.plot(data)
plt.show()
```

8. Time Series Applications

- Stock market prediction
- Weather forecasting
- Sales forecasting
- Demand prediction
- Traffic prediction

9. Interview Questions

1. What is stationarity?
2. Difference between AR and MA?
3. What is ARIMA?
4. What is seasonality?
5. Difference between ACF and PACF?
6. Additive vs Multiplicative model?
7. What is white noise?

10. Real Life Example

Suppose your YouTube subscribers increase monthly:

- Trend → Overall growth
- Seasonality → More subscribers during exam season

- Noise → Viral video spike

If you want, next we can do:

- 🔥 Mathematical derivation of ARIMA
- 🔥 Time Series in Machine Learning
- 🔥 LSTM for Time Series
- 🔥 Full project in Python
- 🔥 Interview preparation deep level

Tell me your level (Beginner / Intermediate / Advanced) 🧙

You said:

with code

ChatGPT said:

Time Series With Code (Beginner → Advanced)

We'll go step-by-step using **Python** (Pandas + Matplotlib + Statsmodels).

◆ 1 Create a Simple Time Series

```
Python
import pandas as pd
import matplotlib.pyplot as plt

# Create date range
dates = pd.date_range(start="2024-01-01", periods=12,
                      freq='M')

# Sample sales data
sales = [100,120,130,150,170,160,180,200,210,230,250,270]

# Create DataFrame
df = pd.DataFrame({"Date": dates, "Sales": sales})
df.set_index("Date", inplace=True)

# Plot
plt.figure()
plt.plot(df["Sales"])
plt.title("Monthly Sales")
plt.show()
```

◆ 2 Check Stationarity (ADF Test)

We use **ADF (Augmented Dickey-Fuller Test)**.

```
Python
from statsmodels.tsa.stattools import adfuller

result = adfuller(df["Sales"])

print("ADF Statistic:", result[0])
print("p-value:", result[1])
```

👉 If **p-value < 0.05** → **Stationary**

◆ 3 Make Data Stationary (Differencing)

```
Python
df["Sales_diff"] = df["Sales"].diff()

plt.figure()
plt.plot(df["Sales_diff"])
plt.title("Differenced Sales")
plt.show()
```

4 ACF and PACF Plot

```
Python
from statsmodels.graphics.tsaplots import plot_acf, plot_pacf

plot_acf(df["Sales_diff"].dropna())
plt.show()

plot_pacf(df["Sales_diff"].dropna())
plt.show()
```

Used to determine:

- $p \rightarrow$ AR order
- $q \rightarrow$ MA order

5 AR Model

```
Python
from statsmodels.tsa.ar_model import AutoReg

model = AutoReg(df["Sales"], lags=1)
model_fit = model.fit()

print(model_fit.summary())
```

6 ARIMA Model

ARIMA(p, d, q)

```
Python
from statsmodels.tsa.arima.model import ARIMA

model = ARIMA(df["Sales"], order=(1,1,1))
model_fit = model.fit()

print(model_fit.summary())

# Forecast next 3 months
forecast = model_fit.forecast(steps=3)
print("Forecast:", forecast)
```


7 SARIMA (Seasonal ARIMA)

```
Python
from statsmodels.tsa.statespace.sarimax import SARIMAX

model = SARIMAX(df["Sales"],
                 order=(1,1,1),
                 seasonal_order=(1,1,1,12))

result = model.fit()

print(result.summary())
```

8 Facebook Prophet Model

Developed by **Facebook** (now **Meta Platforms**)

```
Python
from prophet import Prophet

# Prophet needs specific column names
prophet_df = df.reset_index()
prophet_df.columns = ["ds", "y"]

model = Prophet()
model.fit(prophet_df)

future = model.make_future_dataframe(periods=3, freq='M')
forecast = model.predict(future)

model.plot(forecast)
plt.show()
```

9 LSTM for Time Series (Deep Learning)

Using **Keras**

```
Python
import numpy as np
from tensorflow.keras.models import Sequential
from tensorflow.keras.layers import LSTM, Dense

# Prepare data
data = np.array(sales).reshape(-1,1)

X = []
y = []

for i in range(len(data)-3):
    X.append(data[i:i+3])
    y.append(data[i+3])

X = np.array(X)
y = np.array(y)

# Build model
model = Sequential()
model.add(LSTM(50, activation='relu', input_shape=(3,1)))
model.add(Dense(1))
model.compile(optimizer='adam', loss='mse')

model.fit(X, y, epochs=200, verbose=0)

prediction = model.predict(X)
print(prediction)
```

Summary

Model Use Case

AR Simple past dependency

ARIMA Trend handling

SARIMA Seasonality

Prophet Business forecasting

LSTM Complex patterns

If you want next:

-  Full end-to-end project
-  Mathematical derivation of ARIMA
-  Interview questions with answers

- 🔥 Real stock market example
- 🔥 Machine learning vs classical time series comparison

Tell me your level 🧙

You said:

🔥 9. Interview Questions What is stationarity? Difference between AR and MA? What is ARIMA? What is seasonality? Difference between ACF and PACF? Additive vs Multiplicative model? What is white noise? explain with definition easy way

ChatGPT said:

Perfect 🧙 Let's answer these in **easy interview language** (short, clear, confident style).

🔥 1 What is Stationarity?

✅ Easy Definition:

A time series is **stationary** if its statistical properties do not change over time.

In simple words:

- Mean is constant
- Variance is constant
- No trend or seasonality

Example:

Random daily temperature fluctuation around same average.

Interview Tip:

"Stationarity means the data behavior is stable over time."

🔥 2 Difference Between AR and MA

♦ AR (AutoRegressive)

Current value depends on **past values**.

Example:

Code

$$Y_t = c + \phi_1 Y_{t-1} + \phi_2 Y_{t-2}$$

👉 Uses previous actual values.

◆ MA (Moving Average)

Current value depends on **past errors**.

Example:

Code

$$Y_t = \mu + \theta_1 \varepsilon_{t-1} + \theta_2 \varepsilon_{t-2}$$

👉 Uses previous prediction errors.

🎯 Simple Difference:

AR

MA

Uses past values Uses past errors

PACF cuts off ACF cuts off



3 What is ARIMA?

Full Form:

AutoRegressive Integrated Moving Average

Easy Definition:

ARIMA is a model used to forecast time series data by combining:

- AR (past values)
- I (differencing to remove trend)
- MA (past errors)

Format:

Code

ARIMA(p, d, q)

- $p \rightarrow$ AR order
- $d \rightarrow$ Differencing
- $q \rightarrow$ MA order

Interview Line:

"ARIMA is used for non-stationary time series after making it stationary using differencing."



What is Seasonality?

Easy Definition:

Seasonality is a repeating pattern in fixed time intervals.

Example:

- Ice cream sales increase every summer.
- More shopping during Diwali.

Key Point:

- Fixed period (daily, monthly, yearly)



Difference Between ACF and PACF

◆ ACF (AutoCorrelation Function)

Shows correlation between current value and past values (including indirect effect).

◆ PACF (Partial AutoCorrelation Function)

Shows direct correlation with a specific lag (removes indirect effects).



Simple Difference:

ACF

PACF

Total correlation Direct correlation

Helps find MA(q) Helps find AR(p)

6 Additive vs Multiplicative Model

◆ Additive Model

Used when seasonal variation is constant.

Code

$$Y = T + S + E$$

👉 Example: Sales increase by fixed 100 every year.

◆ Multiplicative Model

Used when seasonal variation increases with trend.

Code

$$Y = T \times S \times E$$

👉 Example: Sales increase by 10% every year.

Difference:

Additive

Multiplicative

Constant variation Increasing variation

Linear growth Percentage growth

7 What is White Noise?

Easy Definition:

White noise is a random series with:

- Mean = 0
- Constant variance
- No autocorrelation

In simple words:

Pure randomness.

Example:

Code

```
5, -2, 3, 0, -1, 4
```

No pattern, no trend.

Interview Line:

"White noise is completely random data with no predictable structure."

Final Quick Revision (30-Second Interview Version)

- Stationarity → Statistical properties constant over time
- AR → Depends on past values
- MA → Depends on past errors
- ARIMA → AR + Differencing + MA
- Seasonality → Repeating fixed pattern
- ACF → Total correlation
- PACF → Direct correlation
- Additive → Constant variation
- Multiplicative → Percentage variation
- White Noise → Pure randomness

If you want, next I can give:

- 🔥 Deep mathematical explanation
- 🔥 Graphical intuition
- 🔥 Tricky interview follow-up questions
- 🔥 Practical examples with Python

Tell me your level 🕶️

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